

**MBA
USP
ESALQ**

ANALYTICS AND RISK MANAGEMENT

Prof. Fabiano Guasti Lima

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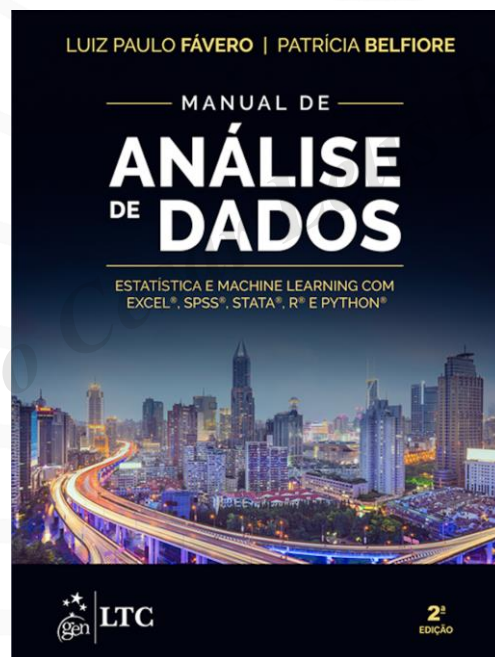
Law No. 9610/98

Program

Market Risk management models, operational and compliance; Basel III; credit risk models; models for other types of applicable risks (legal and regulatory risk; social and environmental risk; reputational); real and practical risks.

Bibliographic references

FAVERO, Luiz Paulo, BELFIORE, Patrícia. **Análise de dados: estatística e modelagem multivariada com Excel, SPSS, Stata, R e Python**. 2. ed. Gen, LTC, 2024.



LIMA, Fabiano Guasti. **Risk Analysis** 3rd ed. São Paulo: Atlas, 2023.



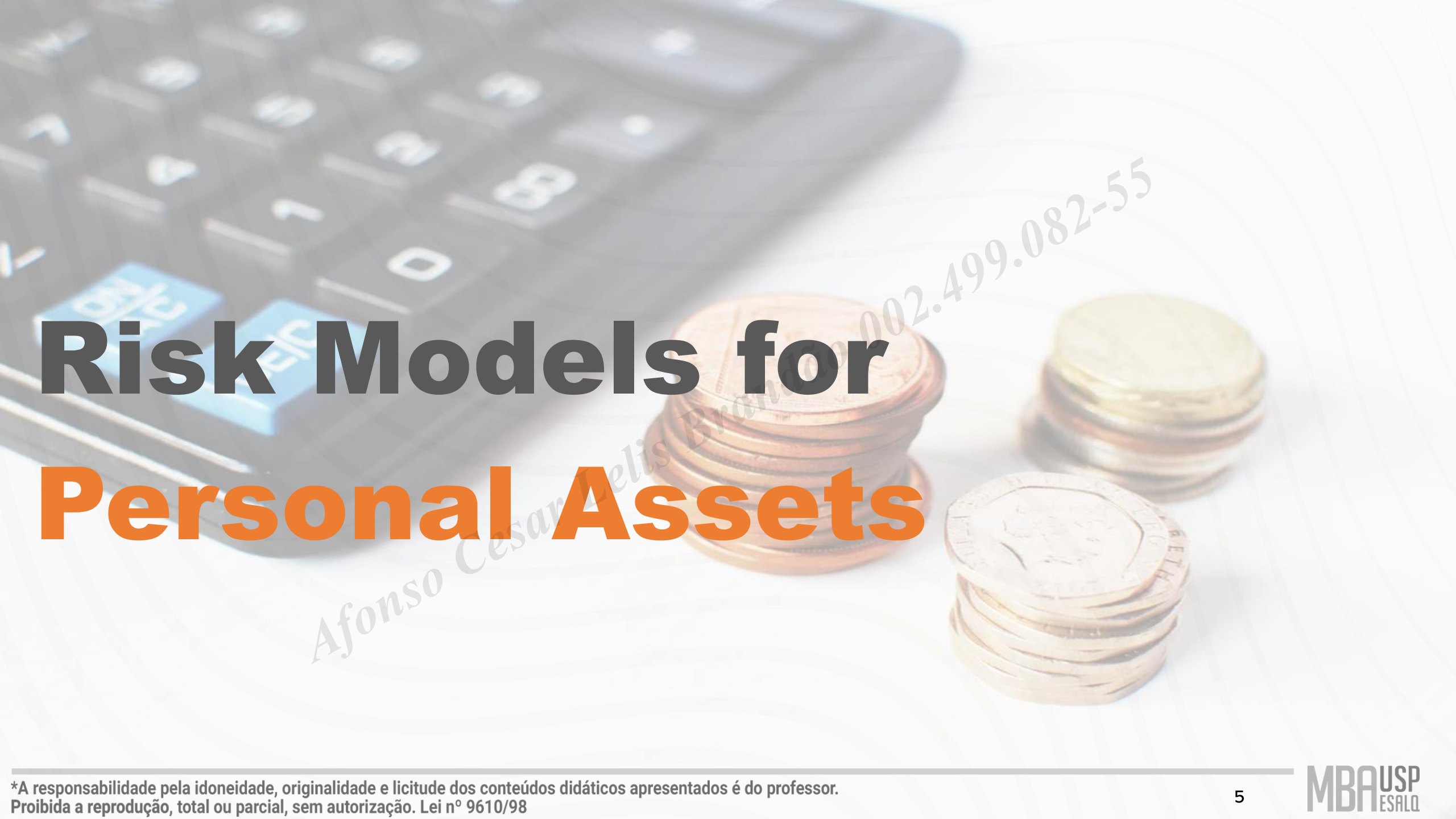
AGENDA

Analytics for the Risk
Models assessment:

- Market
- Credit
- Operational

R Scripts





Risk Models for Personal Assets

Afonso Cesar Pelis Branco 002.499.082-55

Risk and Return of Individual Assets



Ibovespa



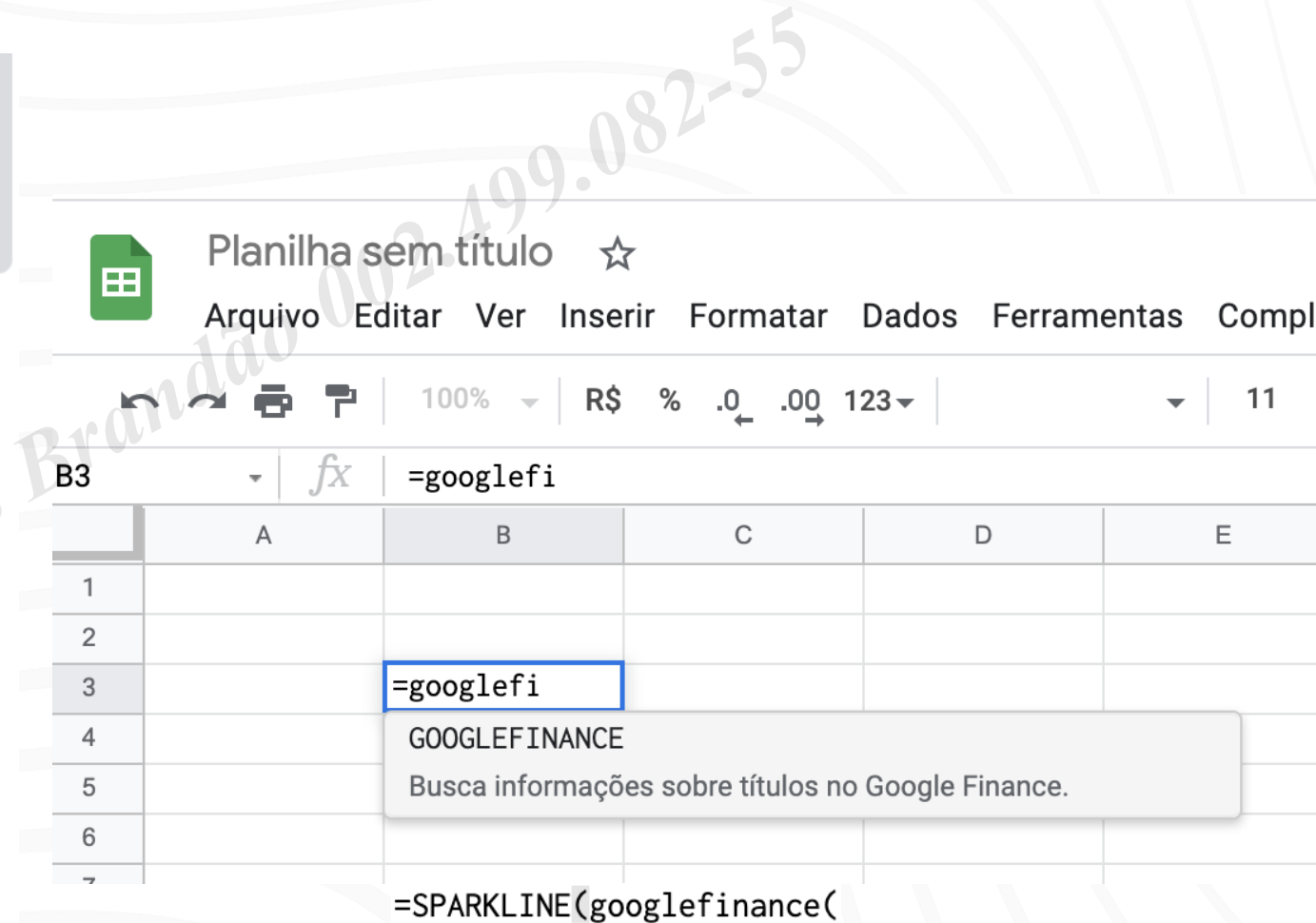
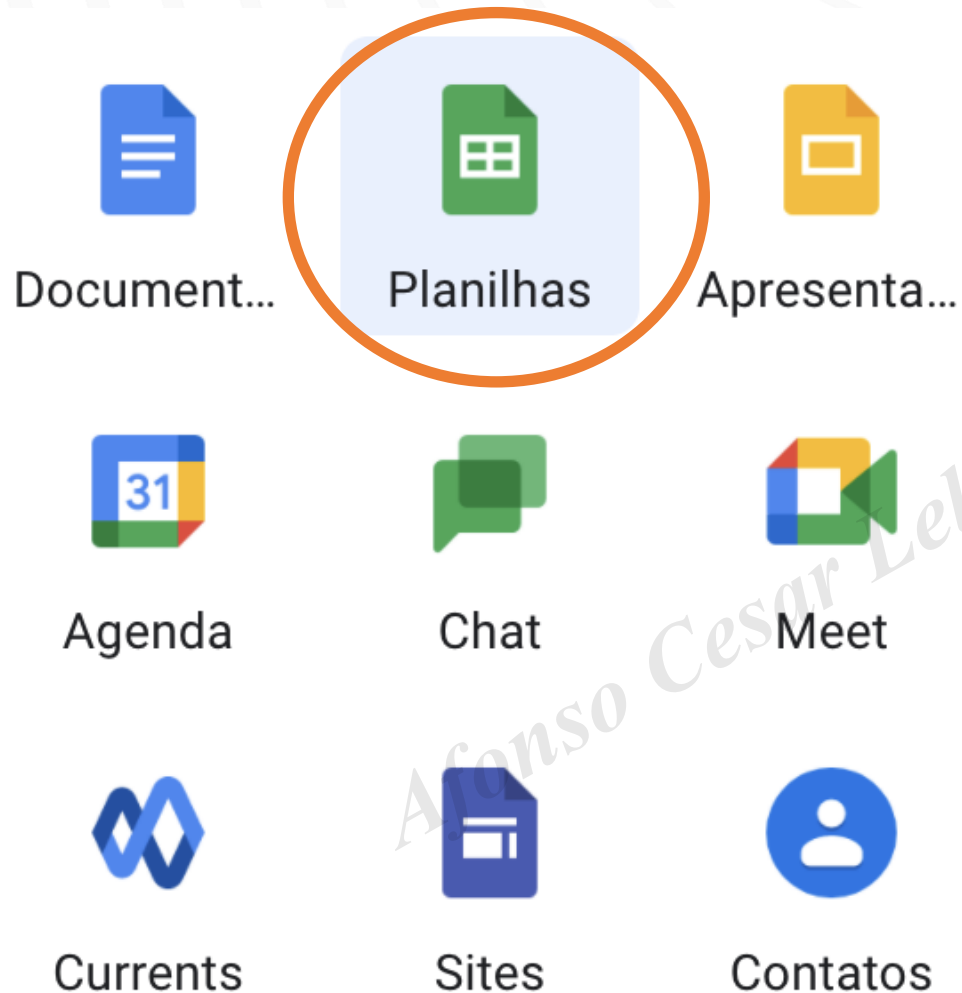
	PETR4	Ibovespa	Café Arábica
Sept/23	R\$ 32.63	117,893	R\$ 818.18
Oct23	R\$ 34.12	115,057	R\$ 797.38
Nov/23	R\$ 35.12	118,160	R\$ 836.59
Dec/23	R\$ 35.67	128,185	R\$ 942.08

Sources: PETR4 and Ibovespa: B3; Café Arábica – CEPEA/Esalq – Images: google images – Access on 01/20/2024



Which one oscillates the most?

Searching for data



Risk and Return of Individual Assets

- Return
 - Discrete

$$R_t = \frac{P_t}{P_{t-1}} - 1 = \frac{\text{Price}_t - \text{Price}_{t-1}}{\text{Price}_{t-1}}$$

- Continuous

$$R_t = \log \frac{P_t}{P_{t-1}}$$

The background of the slide features a blurred image of a calculator on the left and several stacks of coins (copper and gold) on the right. A faint, diagonal watermark text "Afonso Cesar Lelis Brandão 002.499.082-55" is visible across the center of the image.

Risk in the Context of Investment Portfolio

Risk and Return in the context of portfolios

	Asset A	Asset B
Year 1	13.41%	26.62%
Year 2	0.94%	45.13%
Year 3	10.21%	20.21%
Year 4	4.91%	17.37%
Year 5	10.53%	15.67%

	Asset A	Asset B
AVERAGE RETURN	8%	25%
RISK	5%	12%

Harry Markowitz (1927-2023)

Portfolio Selection

Harry Markowitz

The Journal of Finance, Vol. 7, No. 1. (Mar., 1952), pp. 77-91.

Stable URL:

<http://links.jstor.org/sici?sici=0022-1082%28195203%297%3A1%3C77%3APS%3E2.0.CO%3B2-I>

The Journal of Finance is currently published by American Finance Association.



The Journal of Finance – **Portfolio Selection** - Harry Markowitz (1952), v.7, n.1, p. 77-91

https://www.math.ust.hk/~maykwok/courses/ma362/07F/markowitz_JF.pdf

Source: Google Images

Risk and Return in the context of portfolios

	Asset A	Asset B
RETURN	8%	25%
RISK	5%	12%
Investment	R\$ 5,000.00	R\$ 5,000.00
Percentage	50%	50%

$$R_{cart} = 5.000 \times 0,08 + 5.000 \times 0,25 = 400 + 1.250 = 1.650,00$$

$$R_{cart} = 0,50 \times 0,08 + 0,50 \times 0,25 = 0,165 = 16,5\%$$

Risk and Return in the context of portfolios

Asset	Part. %	Return	Risk
A	W_a	R_a	S_a
B	W_b	R_b	S_b
Portfolio	100%	$R_C = W_a \times R_a + W_b \times R_b$	Expression below



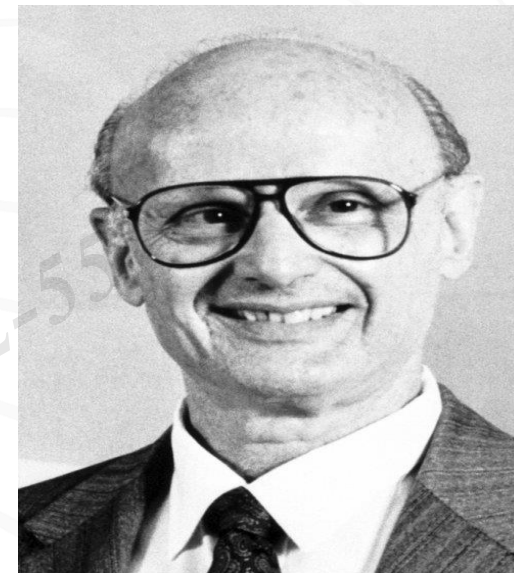
$$S_C = \sqrt{W_a^2 S_a^2 + W_b^2 S_b^2 + 2W_a W_b \underbrace{\text{corr}(R_a, R_b) S_a S_b}_{\text{Covariance } (R_a, R_b)}}$$

or

$$S_C = \sqrt{W_a^2 S_a^2 + W_b^2 S_b^2 + 2W_a W_b \text{cov}(R_a, R_b)}$$

Harry Markowitz (1927-2023)

Video Suggestion:



Source: Google Images

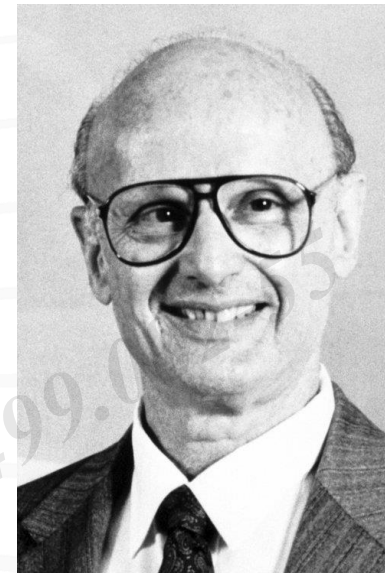
1990 – Nobel Prize in Economic Sciences

Hear What Dr. Harry Markowitz, A Superhero Of Modern Economics, Has To Say About Investing – Entrevista 21/03/2019

<https://www.youtube.com/watch?v=os0B1oQfNdo>

Harry Markowitz (1927-2023)

Video Suggestion:



Source: Google Images

1990 – Nobel Prize
in Economics

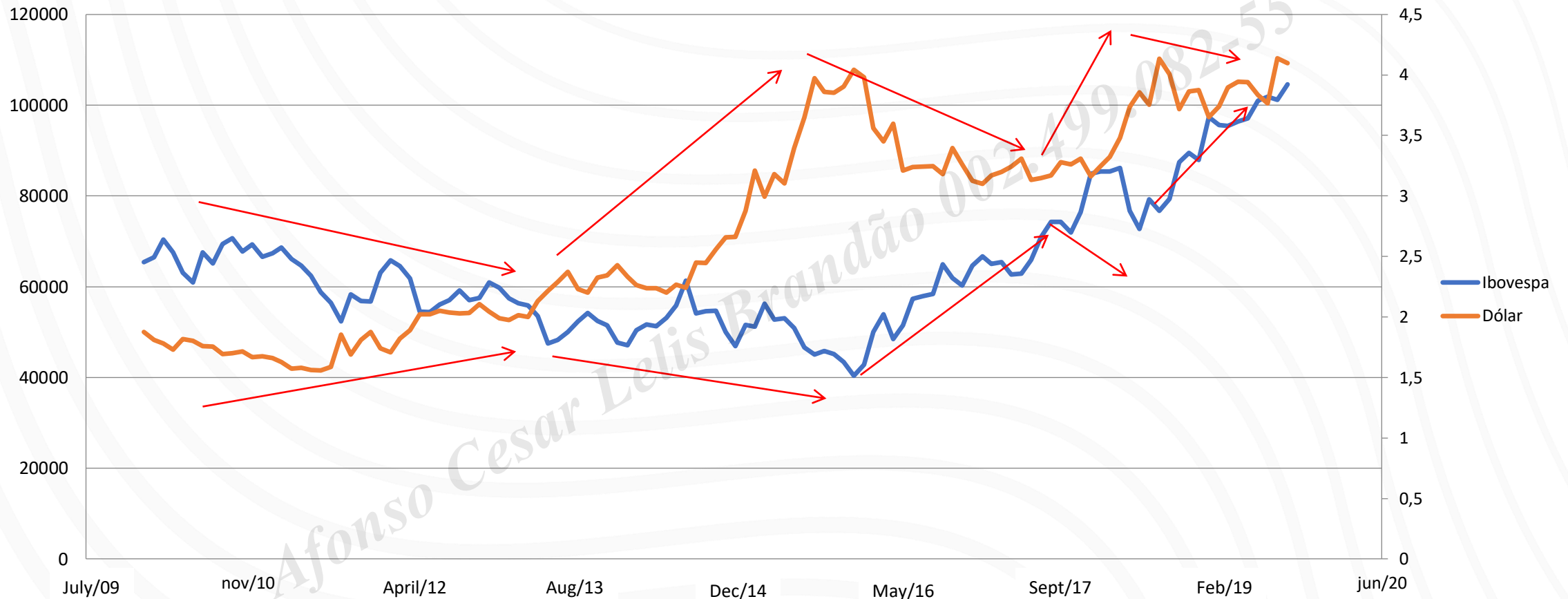
IFA.com with Mark Hebner - An Hour with Harry Markowitz, Father
of Modern Portfolio Theory – Interview - 06/19/2020

<https://www.youtube.com/watch?v=lq2PL-Hxmio>

Correlation Effect

correl = - 0.627

Ibovespa X Dollar

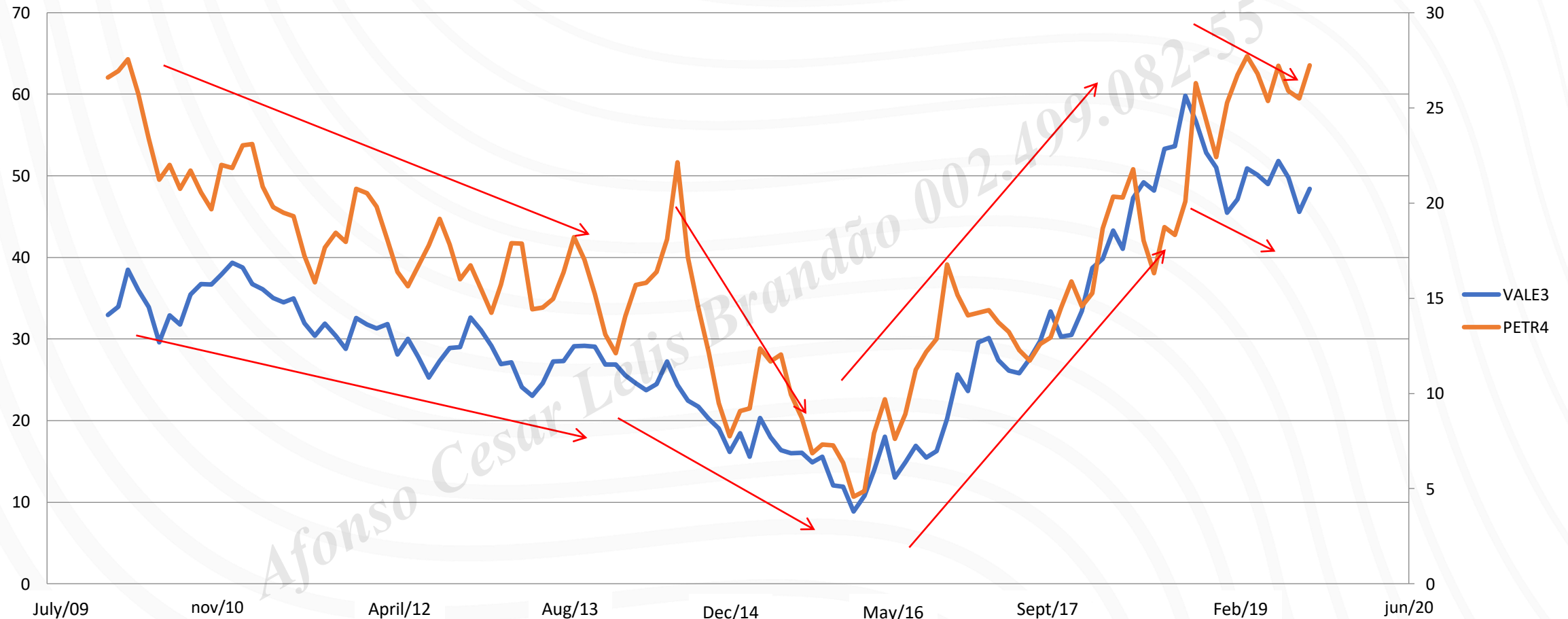


Source: Economática – Jan/2010 to Sep/2019

Correlation Effect

correl = 0.532

PETR4 X VALE3



Source: Economática – Jan/2010 to Sep/2019

Risk and Return in the context of portfolios

	Asset A	Asset B
RETURN	8%	25%
RISK	5%	12%
Investment	R\$ 5,000.00	R\$ 5,000.00
Percentage	50%	50%

correl = - 0.60

$$S_{Port} = \sqrt{(0,50)^2 \cdot (0,05)^2 + (0,50)^2 \cdot (0,12)^2 + 2 \cdot (0,50) \cdot (0,50) \cdot (-0,60) \cdot (0,05) \cdot (0,12)}$$

Portfolio with more than two assets

$$Ret_{Port} = \sum_{i=1}^n W_i \times Ret_{active_i}$$

$$Risk_{Port} = \sqrt{[W_1 \quad \cdots \quad W_n] \times \begin{bmatrix} covar_{11} & \cdots & covar_{1n} \\ \vdots & \ddots & \vdots \\ covar_{n1} & \cdots & covar_{nn} \end{bmatrix} \times \begin{bmatrix} W_1 \\ \vdots \\ W_n \end{bmatrix}}$$

The background of the slide features a blurred image of a calculator on the left and three stacks of coins (copper and gold) on the right. A faint, diagonal watermark text "Afonso Cesar Pelis Brândão 002.499.082-55" is visible across the center of the image.

Models of Credit Risk

Credit Risk

Credit risk is the risk of economic loss due to the non-compliance with contractual obligations by a counterparty

Credit Risk

- Recent history has shown that credit problems are fundamental to the economy, companies, and financial and non-financial institutions...

Asian, Russian, subprime crisis...

Pre-settlement risk

• Accounting

- Loss Incurred – IAS 39
- Probable Loss – COSIF
- Expected Loss – IFRS9

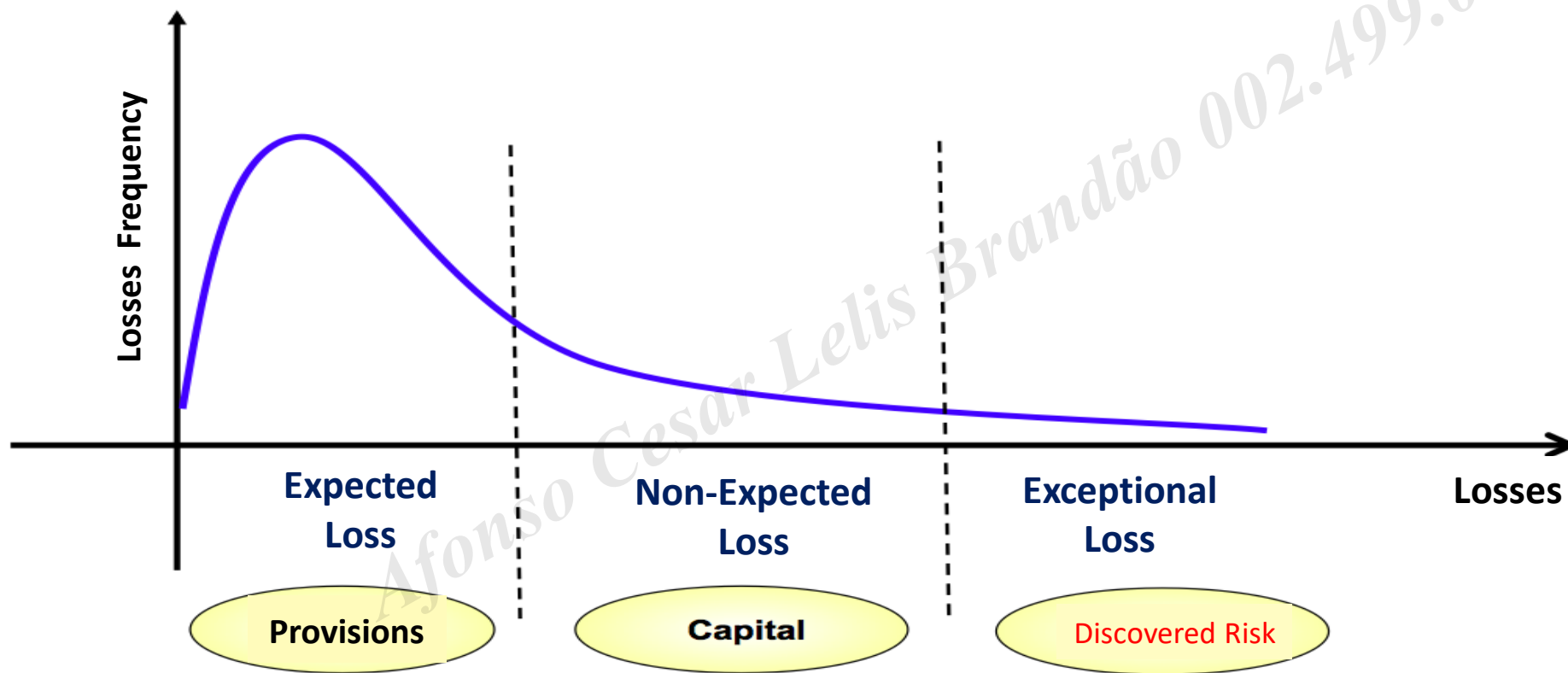
• Prudential/Management

- *Expected Loss* – EL
- *Unexpected Loss* – UL

• **Provision:** supports expected losses

• **Capital:** supports unexpected losses

Credit Risk



Credit Risk

Credit risk drivers

Default (b)

- A counterparty is or is not in default
- This event involves a default probability

Credit exposure at default

- Uncertainty as to the amount of credit exposure at the time of default

Default data loss (event-conditioned) - PDF

- Uncertainty as to the amount that can be recovered after default
- 1 minus the creditor's recovery fee

Credit Risk

$$\text{Credit Loss}_1 = b \times EC \times PD$$

Negotiation with the debtor to recover part or all of the amount owed.

- We will call t the recovery rate
- Therefore, $1 - t$ is the conditioned percentage loss in the default event

$$\text{Credit Loss}_1 = b \times EC \times (1-t)$$

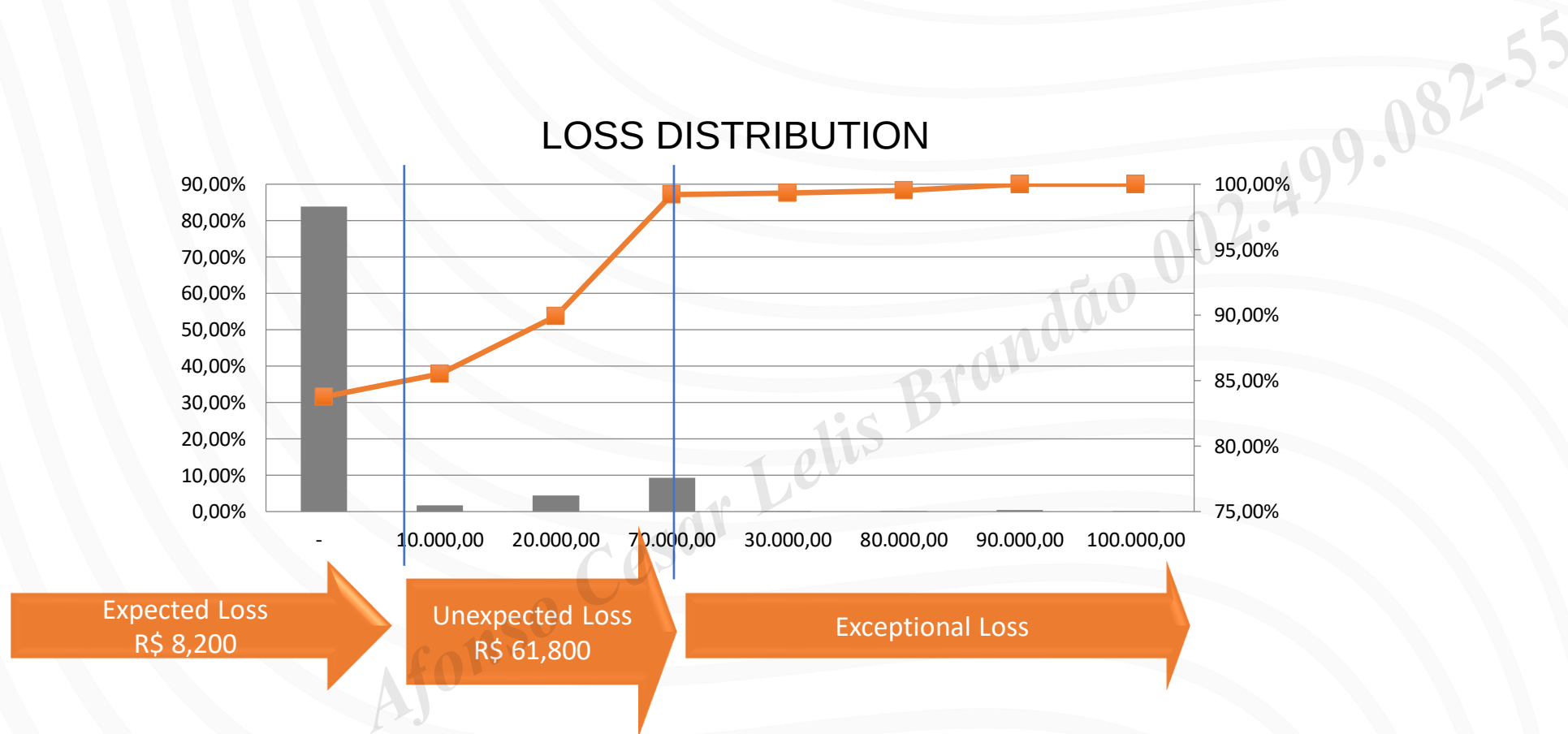
Credit Risk

Consider a portfolio with a total credit exposure of R\$100k invested in three credits.

- Credit A: EC = R\$ 10 thousand; $p = 0.02$
- Credit B: EC = R\$ 20 thousand; $p = 0.05$
- Credit C: EC = R\$ 70 thousand; $p = 0.10$

Assumptions: constant exposures; recovery rate = 0; independent defaults

Credit Risk



Risk Management

Basel Committee – located in the city of Basel (Switzerland) formed by members of Central Banks from developed countries which debate and formalize studies and documents.

Basel I, II, and III – another name for the Capital Agreement, which establishes solvency and leverage indicators for financial institutions.

New Capital Agreement – it is a new compendium of best practices for global risk management (also known as Basel II).

BIS – is a financial institution called the central bank of central banks or in English: Bank for International Settlements

Risk Management

BIS – Bank for International Settlements

```
graph TD; BIS[BIS – Bank for International Settlements] --- P1[Founded in 1930, it is an international organization responsible for banking supervision.]; BIS --- P2[It aims to "promote cooperation between central banks and other agencies in the pursuit of monetary and financial stability".]; BIS --- P3[Headquartered in Basel, Switzerland, it brings together 55 central banks from around the world.];
```

Founded in 1930, it is an international organization responsible for banking supervision.

It aims to "promote cooperation between central banks and other agencies in the pursuit of monetary and financial stability".

Headquartered in Basel, Switzerland, it brings together 55 central banks from around the world.

Risk Management

- **Bank for International Settlements (BIS):**
 - Strengthen international financial cooperation
- **Basel Committee on Banking Supervision (BCBS):**
 - Solvency, liquidity and stability of the international banking system
- **Basel Agreement:**
 - 1988: Basel 1 – capital requirement for credit risk
 - 1996: Capital requirement for market risk
 - 2004: Basel 2 – capital requirement for operational risk
 - 2007: Reference Equity and Required Reference Equity
 - 2009: Basel 2.5 – securitization
 - 2010: Basel 3

Risk Management

The focus of the First Agreement is on credit, that is, institutions had to have sufficient equity capital to guarantee the operations carried out, with a minimum limit set at 8%.

The Current Agreement establishes that the Bank's Capital must be greater than or equal to 8% of the sum of the risks incurred:

$$\frac{\text{Capital}}{\text{Market Risk} + \text{Credit Risk}} \geq 8\%$$

Risk Management

Basel I Weighting Criteria

The weighting of credit-risk assets is mainly a function of the type of exposure:

WEIGHTING	ASSET
0%	Cash
	Credit applications related to countries and their central banks
0%, 20% or 50% (at the discretion of the Local Central Bank)	Credit applications related to non-federal public sector companies
20%	Credit Applications Related to Multilateral Development Banks (MDBs)
50%	Secured Credit Applications for Residential Property
100%	Bank-Related Credit Applications
	Business-Related Credit Applications
	All other assets

Risk Management

Basel Index

Asset Weighted by Risk

Example:

Fixed Assets	\$
Cash	1,000
Investments	50,000
Home Loan	200,000
Loans	300,000

Risk Management

Basel Index

Asset Weighted by Risk

Example:

Fixed Assets	\$	Weights	APR - \$
Cash	1,000	0%	0
Investments	50,000	20%	10,000
Real Estate Financing	200,000	50%	100,000
Loans	300,000	100%	300,000
TOTAL	551,000		410,000

Risk Management

Basel Index

Asset Weighted by Risk

Level 1: Share Capital + Capital Reserves + Profit Reserves
+/- Accumulated Profits or Losses + Shareholdings

Level 2: Revaluation Reserves + Contingency Reserves +
Subordinated Debt + Hybrid Capital Instruments

Example:

Net Worth	\$
Level 1	40.000
Level 2	30,000
TOTAL	70,000

Risk Management

Basel Index

Example:

Index	
APR	410,000
PR	70,000
PR/APR	17.1%



Appropriate Institution

Basel III

PATRIMÔNIO DE REFERENCIA (PR)	PATRIMÔNIO DE REFERENCIA MÍNIMO REQUERIDO (PRMR)
• Nível I • Capital Principal = Patrimônio Líquido + Instrumento Elegível a Capital Principal (-) Ajustes Prudenciais • Capital Complementar = Instrumentos Híbridos e Capital e Dívida (inclui uma gama de instrumentos que combinam características passivas e de patrimônio) • Nível II Dívida subordinada elegível a capital	• $f \times RWA$ (ATIVOS PONDERADOS PELO RISCO) • $F =$ • 01.10.2013 a 31.12.2015 = 11% • 01.01.2016 a 31.12.2016 = 9,875% • 01.01.2017 a 31.12.2017 = 9,25% • 01.01.2018 a 21.12.2018 = 8,625% • A partir de 01.01.2019 = 8%

Source: Bacen – presentation of Basel III - 2013



Models of Operational Risk

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Operational Risk Measurement

- Based on the distribution of losses (frequency and severity).
- Use historical data of frequency and operational losses values.
- Use subjective estimates of probabilities and/or values related to frequencies and severities.
- Use theoretical distributions made based on historical data or expertise of the manager.

Operational Risk

Being **n** the number of events related to operational risk within an arbitrated interval. (E.g. 1 year)

Being **S** the severity (magnitude) of each loss considering that it occurred.

Consider the **Probabilities** associated with each of the values of **n** and **S**.

The results are charted by elaborating an empirical distribution of the combinations of events with the respective magnitude

Example:

Number of Loss	Probability
None	70%
1	20%
2	10%

Loss Magnitude	Probability
\$ 500	60%
\$ 10,000	30%
\$ 50,000	10%

Example:

Next steps:

- Combine the possibilities to generate an empirical distribution for operational potential losses
- **Operational VaR**
 - Calculate the **VALUE AT RISK** for the desired level of confidence
- **Operational VaR = WORST LOSS – expected average loss**



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Cash Flow AND ITS RISKS

CASE FLOWS AND INVESTMENT ANALYSIS

Investment is evaluated based on expected future benefits:
CASH FLOWS

TEMPORAL Distribution of cash flows

Effective operating of cash flows, net of income tax. Non-expenditure disbursement

MAIN COMPONENTS OF CASH FLOW

Initial investment



Operational cash inflows



Residual cash flow



CALCULATION OF OPERATING CASH INFLOW

REVENUE

- OPERATIONAL EXPENSES

= **EBITDA**

- DEPRECIATION

= ***EARNINGS BEFORE INTEREST AND TAXES*** = EBIT

- INCOME TAX

= ***NET OPERATING PROFIT AFTER TAX*** = NOPAT

+ DEPRECIATION

= ***OPERATING CASH INFLOW***

Example

- **Investment: R\$ 200,000.00**

- **Revenue:**

(one thousand)	Year 1	Year 2	Year 3	Year 4
Revenue	170	145	125	100

- **Operational Expenses:**

(one thousand)	Year 1	Year 2	Year 3	Year 4
Expenses Op.	23	23	20	20

- **Annual Depreciation: R\$ 50,000.00**

- **Income Tax Rate = 40%**

- **Expected Return Rate = 14% per year**

Suggestion for Simulations:

Palisade is now Lumivero

My Account

LUMIVERO Products Plans Resources Support Company Contact

Try it Free Buy Now

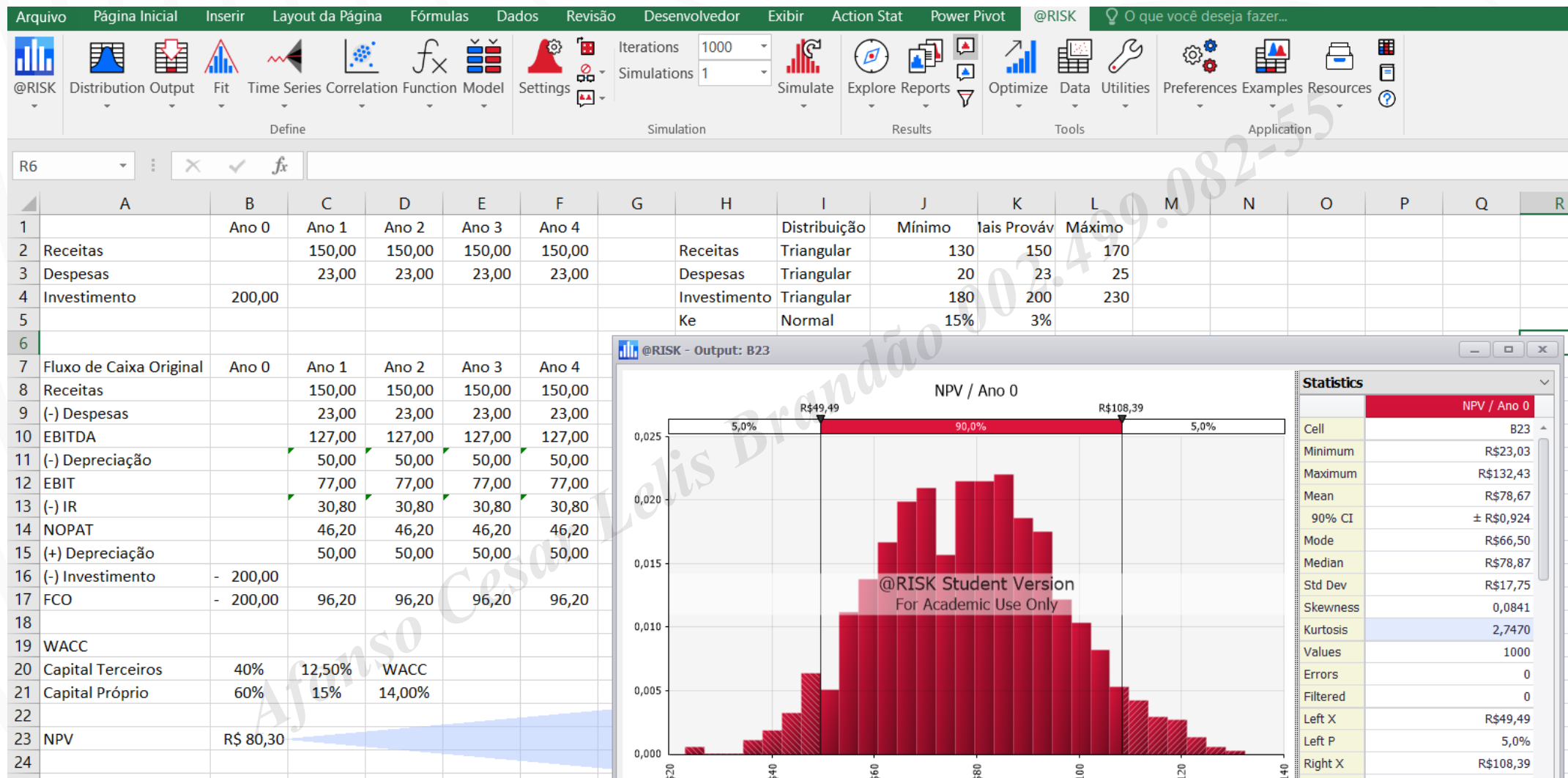
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Better Decisions Start Here

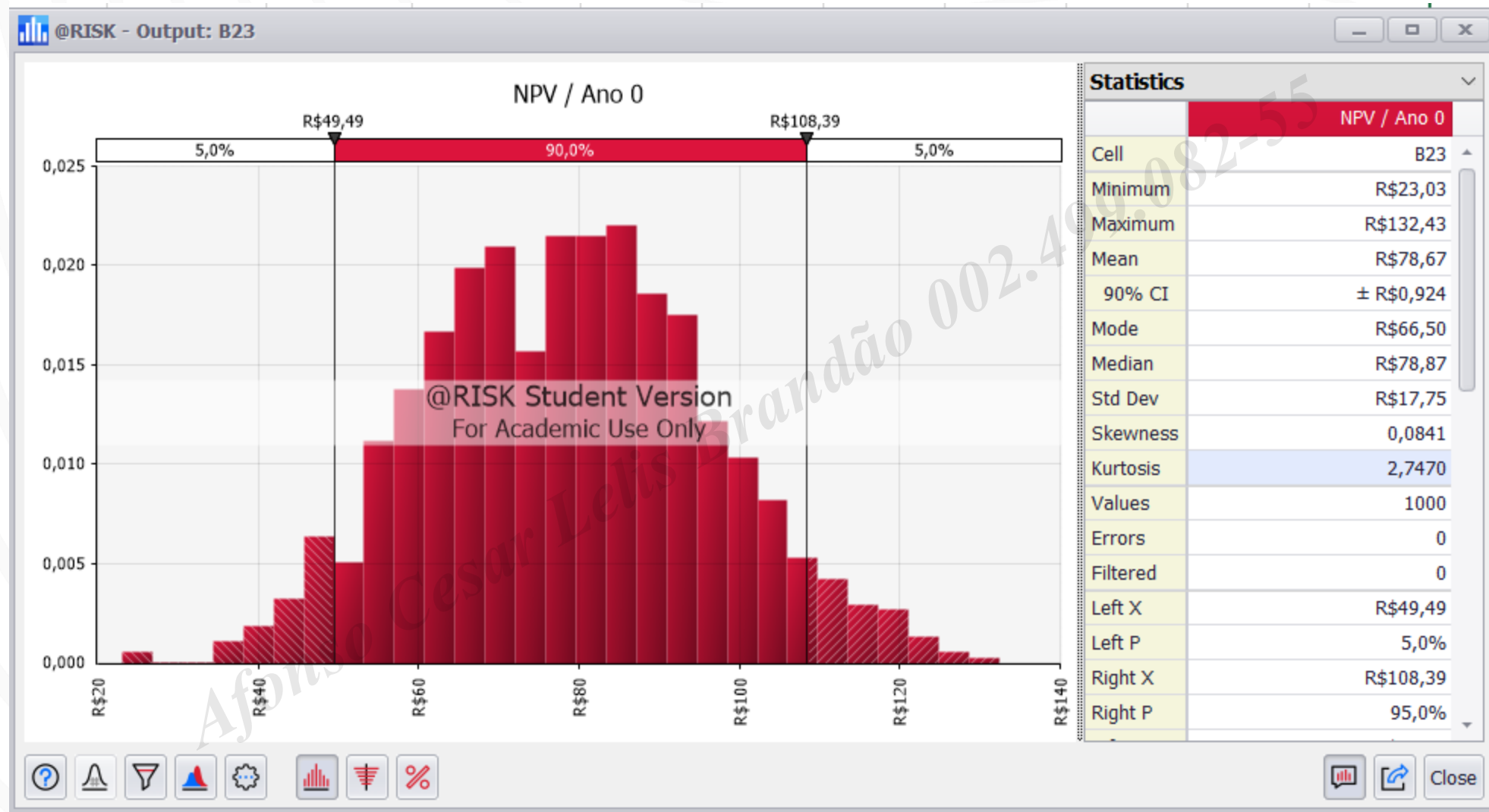
See Why Top Companies Rely on @RISK and DecisionTools Suite

<https://lumivero.com/products/at-risk/>

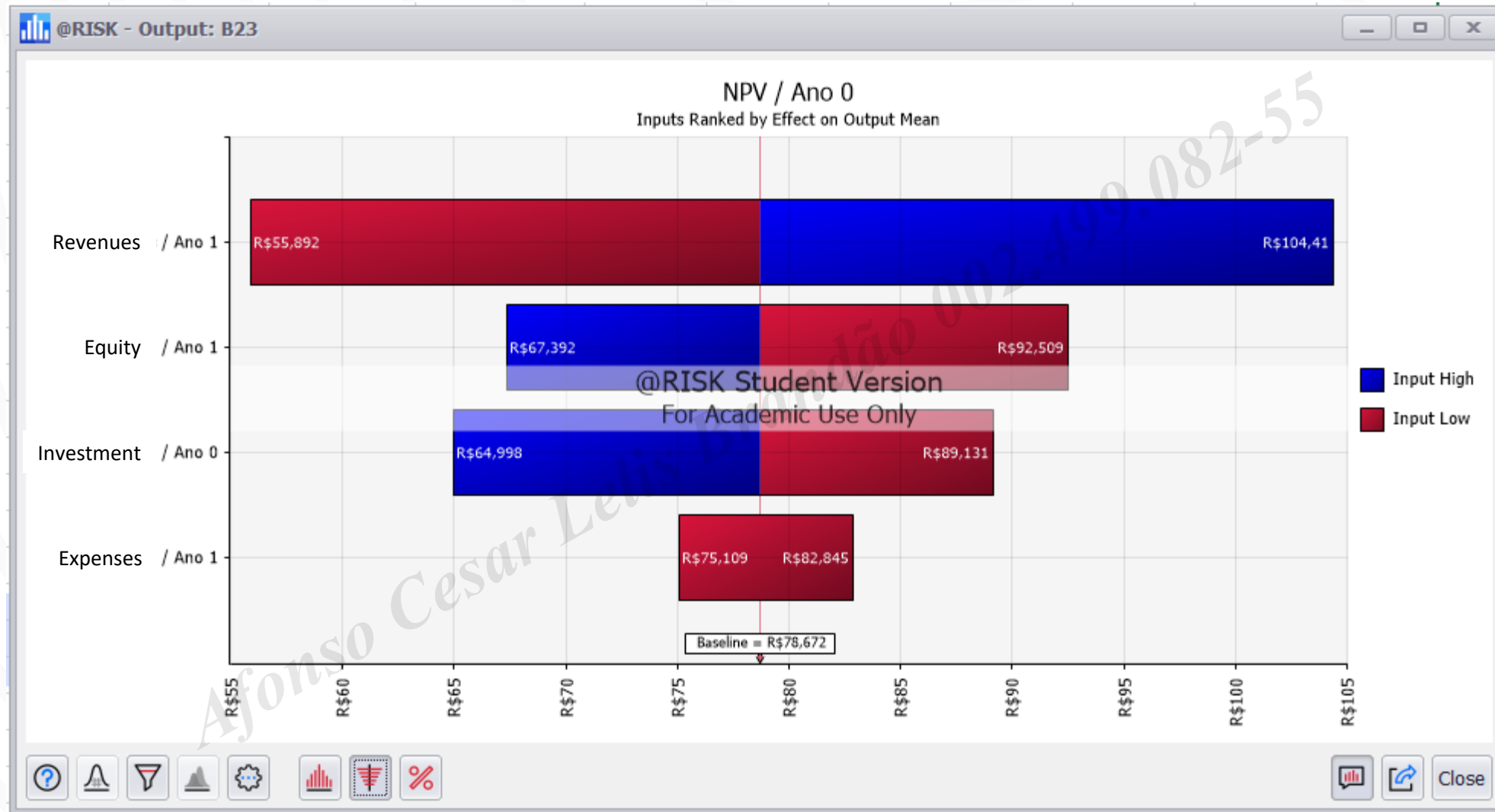
The suggestion of @RISK as an analysis tool in this class has no financial relationship with LUMIVERO. We suggest its use only for the quality of the tool and its advanced analysis capacity.



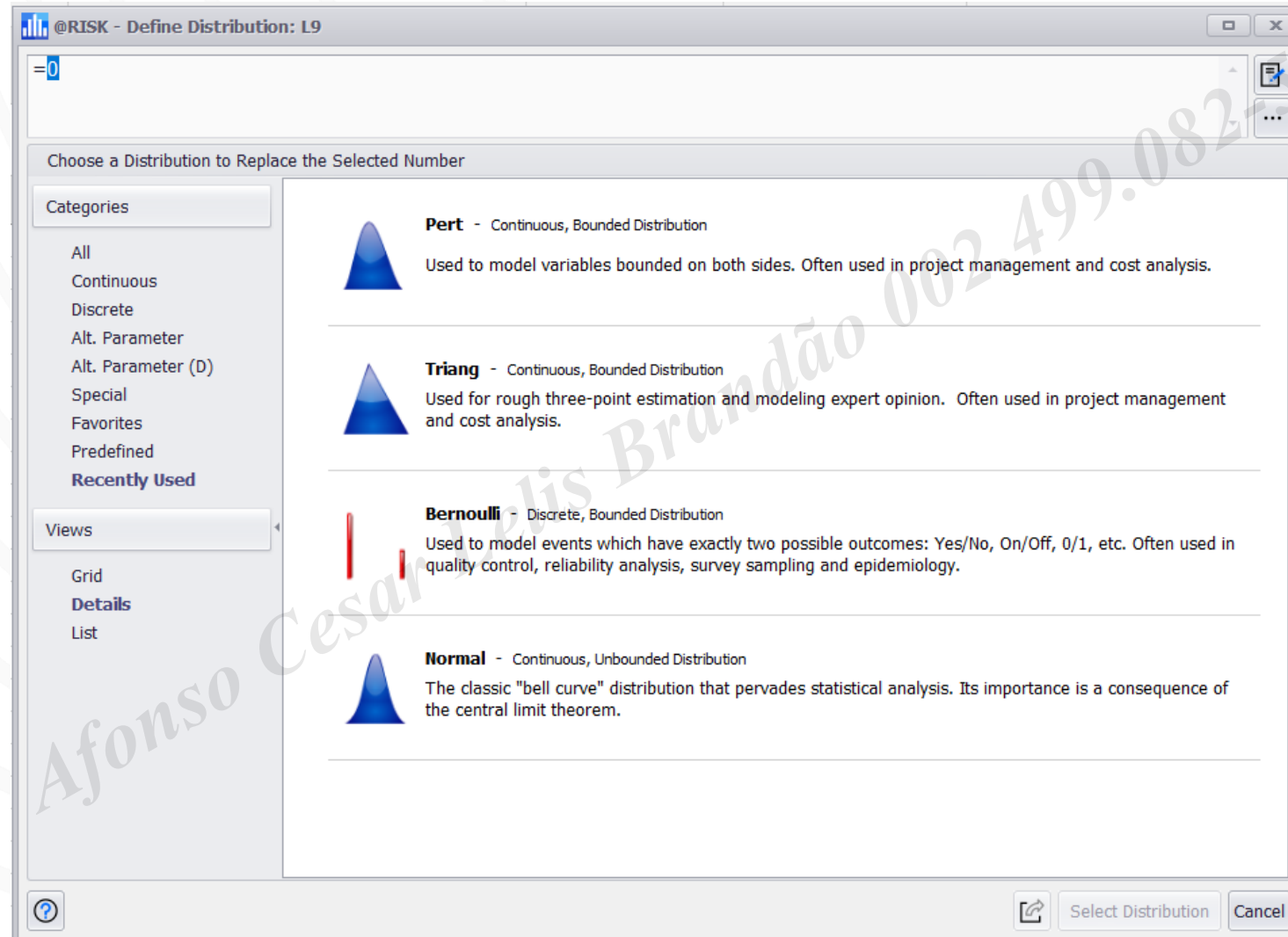
Source: Image of the Worksheet done in class



Source: Image of the Worksheet done in class



Types of Distribution



THANK YOU!

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